ASEAN GEOAI FUSION 2026 · TEAM BRIEFING DOCUMENT

SUNTOWER-AI

Satellite-driven priority ranking for solarising ASEAN's off-grid, diesel-dependent telecom towers.

Challenge theme: Sustainability | **Team:** HACK_ID_068
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> This document is for briefing the team before the hackathon: the problem, the data, the method, the build plan, and the roles — everything needed to get straight into building on the day.

0 Executive Summary

Thousands of telecom towers across ASEAN's off-grid and weak-grid regions still run on diesel generators — expensive, high-carbon, and exposed to fuel-price shocks. Operators are beginning to solarise these sites (PLDT/Smart with EdgePoint Towers in the Philippines; EdgePoint's first hybrid site in Malaysia), but which tower gets solarised first is currently decided ad hoc, not from data. **SUNTOWER-AI** is a satellite-data-driven platform that scores and ranks every off-grid tower in a region by its combined solar yield potential, diesel cost avoided, emissions avoided, and value to underserved communities — so that limited solarisation budgets go to the towers that matter most first.

> **Why this fits the brief:** it sits squarely in the hackathon's **Sustainability** theme (resilient infrastructure, green infrastructure), it is built almost entirely from free satellite imagery, and it solves a real, current, ASEAN-wide telecom infrastructure problem — not just a general environmental one.

1 The Problem

- Thousands of ASEAN telecom towers in remote, off-grid areas still run on costly, high-carbon diesel generators.
- 2026 diesel price shocks (tied to the Iran conflict) are pushing operators toward solar retrofit faster than before.
- Operators are already moving — PLDT/Smart + EdgePoint Towers are solarising 20 off-grid sites in the Philippines, and EdgePoint has deployed its first solar-hybrid site in Malaysia — but site selection is largely manual and opportunistic, not ranked by actual impact.
- There is no systematic, satellite-based way to answer: "*of our 5,000 off-grid towers, which 50 should we solarise this year for maximum benefit?*"

> **Result:** limited capital gets spent on lower-impact sites, while towers with the best solar potential, highest diesel cost, and greatest connectivity value are missed. **Success looks like:** a transparent, data-driven ranking that directs solarisation capital to the towers with the highest combined cost, emissions, and connectivity impact.

2 Who This Is For

Telecom tower companies
Capital allocation decisions for solar retrofit programmes.

National telecom regulators
Sustainability targets and universal service obligation (USO) tracking.

Off-grid rural communities
The people who lose connectivity when diesel supply is delayed or unaffordable.

Renewable energy EPC contractors
Deployment partners who install and maintain solar-hybrid retrofits.

Climate / environmental agencies
Emissions reporting and climate-finance eligibility.

ASEAN Digital Masterplan stakeholders
Regional connectivity & sustainability policy alignment.

3 Where This Applies

Area of interest: off-grid / weak-grid rural regions across ASEAN — outer-island Indonesia, Mindanao (Philippines), upland Myanmar and Laos are the clearest examples, but the method generalises to any ASEAN member state with a documented tower inventory.

Scale: ASEAN region → Country → Province → individual tower site (the model scores *every single tower*, not just a region — this is what makes it granular and actionable).

Key spatial layers: solar irradiance, land cover / shading obstructions, terrain and slope, nighttime-lights (grid proximity), and population / connectivity-gap zones.

4 Data Sources

Every layer below is free, satellite-derived, and accessible through Google Earth Engine (GEE). Currency was verified directly against the live GEE Data Catalog in August 2026 — two commonly-used datasets turned out to be stale, so current alternatives are used instead.

Purpose	Dataset	GEE snippet	Status	Notes
Solar yield (baseline)	Global Solar Atlas PVOUT/GHI	`projects/sat-io/open-datasets/global_solar_atlas/pvout_LTAy_AvgDailyTotals`	BASELINE	Long-term climatological average — the correct choice for a yield estimate, not meant to update yearly.
Solar yield (recent x-check)	MODIS Downward Shortwave Radiation	`MODIS/062/MCD18A1`	LIVE	Updated through Mar 2026.

Land cover / shading	Dynamic World V1	`GOOGLE/DYNAMICWORLD/V1`	LIVE	10m, near-real-time, updates every 2–5 days. Replaces ESA WorldCover v200, which is a static 2021 snapshot.
All-weather land monitoring	Sentinel-1 SAR GRD	`COPERNICUS/S1_GRD`	LIVE	Daily, radar — sees through cloud cover, important for ASEAN's monsoon skies.
Terrain / shading geometry	Copernicus DEM GLO-30 (2024_1)	`COPERNICUS/DEM/GLO30_2024_1`	CURRENT	Surface model (includes buildings/vegetation height) — more realistic for shading near a tower than bare-earth SRTM.
Off-grid confirmation	VIIRS Nighttime Lights (monthly)	`NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG`	LIVE	Updated through Feb 2026. Low/zero radiance near a tower = genuinely off-grid.
Connectivity-gap weighting	GHSL Population Grid P2023A	`JRC/GHSL/P2023A/GHS_POP`	CURRENT	Multitemporal 1975–2030. Replaces the official WorldPop 100m layer, which stops at 2021.
Site verification imagery	Sentinel-2 Surface Reflectance	`COPERNICUS/S2_SR_HARMONIZED`	LIVE	~5-day revisit, 10–20m.
Access / logistics cost	OpenStreetMap roads	via `geemap.osm_to_ee()` or community FeatureCollection	LIVE	Crowd-sourced, continuously updated — proxy for install/maintenance access cost.

Non-spatial data (operator-provided)

- Tower site list with coordinates, current power source, and diesel consumption/cost records.
- Tower operating expenditure (opex) history.
- Grid-connection status and any existing solarisation pipeline/plans.

5 GeoAI Intelligence

GeoAI task	What it produces
Per-site solar-yield estimation	Expected kWh/day for a standard panel array at each tower, corrected for local shading and terrain.
Diesel-cost-avoidance modelling	Estimated annual $ saved per site if solarised, from current diesel consumption records.
Off-grid / connectivity-gap overlay scoring	How many people in the surrounding underserved area depend on this specific tower.
Multi-criteria site ranking	A single weighted priority score per tower combining all of the above (AHP-style weighting or a trained XGBoost ranker).

Outputs: a ranked tower priority list, a solar-ROI map, an emissions-avoided estimate per site, and a connectivity-impact score.

6 The Solution: SUNTOWER-AI Platform

What it is: a satellite-data-driven prioritisation platform for telecom tower solarisation.

Key modules

- **Solar-yield & shading calculator** — per-site expected energy output.
- **Diesel-cost / ROI dashboard** — payback period and savings per site.
- **Connectivity-equity overlay** — visualises which towers serve the most underserved people.
- **Exportable priority list** — a ranked CSV/table an operator can hand straight to their capex planning team.

7 Technology Stack

Layer	Tools
GIS / satellite processing	QGIS, Google Earth Engine
Cloud	GCP / AWS
AI / ML	Python, XGBoost (multi-criteria ranking model)
Data platform	PostgreSQL / PostGIS
Dashboard	Power BI or React-Leaflet

8 Responsible AI: Ethics, Privacy & Sustainability

- Use aggregated, operator-level cost data only — avoid pulling in sensitive commercial data beyond what the ranking needs.
- Publish the ranking methodology and weightings transparently, so operators and regulators can audit why a tower was ranked where it was.
- Guard against a purely ROI-driven ranking systematically deprioritising the hardest-to-reach — often the poorest — communities; the connectivity-equity weighting exists specifically to counteract this.
- Aligned with SDG 7 (Affordable & Clean Energy), SDG 9 (Industry, Innovation & Infrastructure), and SDG 13 (Climate Action).

9 Expected Value & Impact

Dimension	Impact
Economic	Diesel cost avoided; lower exposure to fuel-price shocks.
Social	More reliable connectivity in off-grid communities (fewer outages from delayed fuel delivery).
Environmental	Measurable CO_2 avoided per solarised site.
KPIs	$ diesel avoided/year · tCO_2 avoided · number of priority sites solarised in Year 1.

10 Workflow / Pipeline

How raw satellite data becomes a ranked, actionable priority list:

Stage 1 — Data Ingestion (Google Earth Engine)

Pull Global Solar Atlas, Dynamic World, Copernicus DEM, VIIRS nighttime lights, GHSL population, and Sentinel-1/2 for the pilot AOI. Clip everything to the operator's tower coordinate list.

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Stage 2 — Feature Engineering (GEE + Python)

Per tower, compute: solar-yield score (irradiance corrected for terrain shading), off-grid confirmation (nighttime-light threshold), land-availability score, and population served / connectivity-gap score.

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Stage 3 — Multi-Criteria Ranking Model (Python)

Combine solar yield + diesel-cost-avoided + emissions-avoided + connectivity impact into one weighted priority score per tower (AHP-style weighted sum, or a trained XGBoost ranker if enough historical solarisation outcomes are available).

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Stage 4 — Dashboard & Output

Interactive map (React-Leaflet or Power BI) showing every tower colour-coded by priority score, plus an exportable ranked list with ROI and emissions-avoided figures per site.

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Stage 5 — Decision

Operator or regulator allocates the year's solarisation budget starting from the top of the ranked list, with the connectivity-equity weighting ensuring underserved communities aren't systematically skipped.

11 Roadmap

Long-term (post-hackathon)

Phase	Focus
Phase 1 (0–3 mo)	Pilot in one off-grid region using open data + a partner operator's tower list.
Phase 2 (3–6 mo)	Build & validate the ranking model against known solarised vs. diesel sites.
Phase 3 (6–9 mo)	Pilot the dashboard with one operator/regulator for real prioritisation decisions.
Phase 4 (9–15 mo)	Scale to multiple ASEAN countries/operators.
Phase 5 (ongoing)	Refine the model with post-solarisation performance data.

Hackathon Day (1-day build plan)

09:00–09:30	Kickoff & scoping — confirm pilot AOI, finalise data sources, assign roles.
09:30–11:00	Data acquisition — pull solar, land cover, DEM, nighttime-lights and population layers from GEE for the pilot AOI; prepare a working tower list.
11:00–13:00	Core GeoAI build — solar-yield scoring, off-grid confirmation, connectivity-gap weighting, multi-criteria ranking.
13:00–13:30	Break
13:30–15:00	Prototype dashboard — interactive map (ranked towers, priority scores, supporting layers).
15:00–16:00	Integration & testing — connect model output to dashboard, validate against reference sites, fix bugs.
16:00–16:45	Responsible-AI check — fairness & transparency pass on the ranking methodology.
16:45–17:30	Pitch deck & demo script preparation.
17:30–18:00	Final rehearsal & submission.

12 Team Roles

- **GeoAI / Data Lead** — owns the GEE data pipeline and the ranking model (Stages 1–3).
- **Dashboard / Frontend Lead** — owns the map and priority-list UI (Stage 4).
- **Ethics & Pitch Lead** — owns the Responsible-AI check, pitch deck, and demo narrative (Stage 5 framing).

Roles can be merged for smaller teams — the stages above are designed to be worked on largely in parallel once the data pull (Stage 1) is done.

13 References

GIS / multi-criteria site-selection methodology

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4. Nema, P., Nema, R.K., & Rangnekar, S. (2010). *Minimization of greenhouse gases emission by using hybrid energy system for telephony base station site application.* Renewable and Sustainable Energy Reviews, 14(6), 1635–1639.
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